

### 3.7 Pythagoras' theorem and trigonometry

| Subject content                                                                             | Guidance                                                                                                                      | Curriculum reference |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>A</b> Identify and name the hypotenuse, adjacent and opposite of a right-angled triangle |                                                                                                                               | G9.5A                |
| <b>B</b> Know and use Pythagoras' theorem in 2-D                                            | $a^2 + b^2 = c^2$<br>Find a missing side given the lengths of the other 2 sides                                               | G9.5B                |
| <b>C</b> Use trigonometry to calculate lengths and angles in a right-angled triangle        | Find a missing side given the length of another side and an angle.<br><br>Find a missing angle given the length of two sides. | G9.5C<br>G9.5E       |
| <b>D</b> Solve problems in context involving right-angled triangles                         | A ladder reaches 5 m up a wall<br><br>The bottom of the ladder is 2 m away from the wall.<br><br>How long is the ladder?      | G9.5D                |